

$$\lim \frac{P_{LL}}{P_{LH} + P_{HL}} \quad \text{and} \quad \lim \frac{P_{LL} + P_{HH}}{P_{LH} + P_{HL}} \quad (13)$$

equal zero when certain parameters tend to infinity in some well-defined way. However, it is worthwhile at this stage to note their implications for cosmology.

The strong broken symmetry in Definition 5 actually means that a monotonic behavior of the entropy is far more probable than a non-monotonic one. In the case of a weak broken symmetry, this is not necessarily so; it could very well be that the most probable scenario would be high entropy at both ends. Thus, this is definitely a weaker statement, but it can nevertheless be argued that it can be used to explain the time asymmetry that we observe, referring to a kind of anthropic principle: it is an obvious observational fact that we live in a universe with low entropy at at least one end. If the statement in Definition 4 is fulfilled, then clearly among such scenarios, the monotonic ones (LH and HL) are the by far most probable ones. Thus, since universes with high entropy at both ends would seem to be quite uninhabitable, one can argue that given the existence of an observer, then with almost certainty he must live in a universe with monotonic entropy.

Summing up, both limits above can be used to argue in favor of time asymmetry. Nevertheless, at least to the mind of the author, the strong broken symmetry is the preferable one. This alternative will be further studied in Section 9.

8. Numerical computations in the combinatorial multiverse

With the setup in Sections 6 and 7, we can now use Mathematica or MATLAB to generate instances of the combinatorial multiverse for small values of m and W and then compute the corresponding probability weights P_{LL} , P_{LH} , P_{HL} and P_{HH} . It is important to note that the matrices here can be treated as sparse, rather than as full matrices, which make the computations considerably faster.

In particular, in the case $m = 2$ in Section 6 and with a randomly generated dynamics which is manifested by an adjacency matrix A , we can compute the power A^4 and read of the first row, which contains all the information we need about the paths from the state at $t = -2$ with $S = 0$. So what do we find?

In **Figure 3**, I have plotted the ratio $N_{LL}/(N_{LH} + N_{HL})$ for the cases $m = 2$ (light gray) and $m = 3$ (dark gray) for values of W ranging from 3 to 30. What is actually displayed are the mean values of 1000 randomly generated matrices as above for each value of W . Although the picture clearly supports the claim that

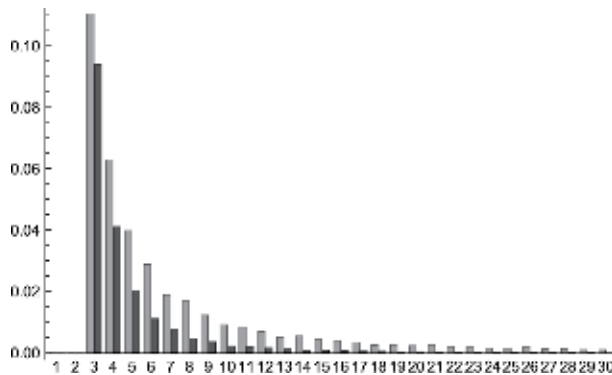


Figure 3. The ratio $N_{LL}/(N_{LH} + N_{HL})$ as a function of W for the cases $m = 2$ (light gray) and $m = 3$ (dark gray) [4].